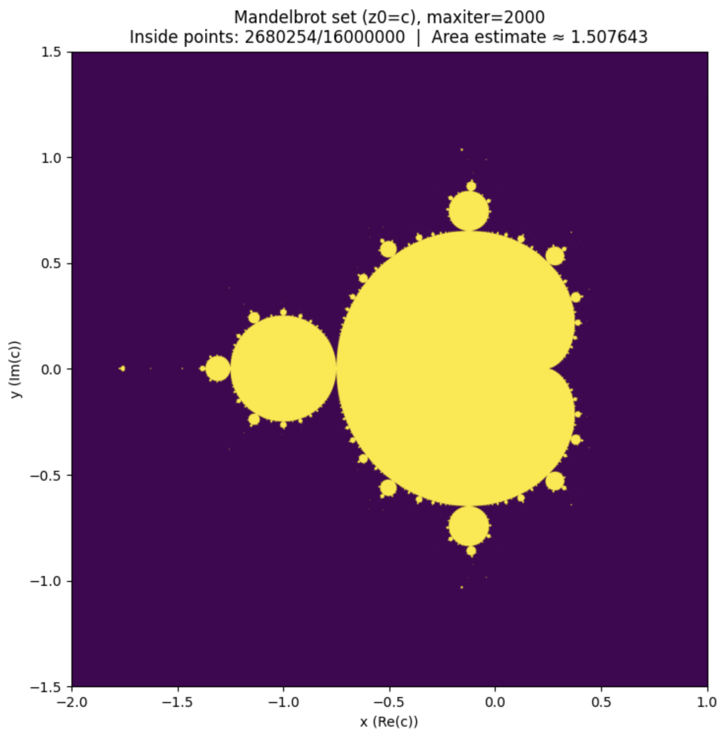
COMPUTATIONAL TOOLS - OpenMP

HOMEWORK: Area of the Mandelbrot Set



The Mandelbrot set is the set of complex numbers $c = x+iy$ for which the iterative sequence $z_{k+1} = z_k^2 + c$ when started from $z_0 = c$ does not diverge. Numerically, the condition is satisfied if $|z_k| \leq 2$ up to a number of `maxiter` steps. The Mandelbrot is a fractal set in the complex plane and is symmetric about the real axis.

In this exercise, you will estimate its area numerically using a grid-based sampling method. There is no known closed-form theoretical value for the area. Choose a rectangular box $[-2, 1] \times [0, 1.5]$ (upper half) and create a grid of $N \times N = 4096 \times 4096$ sample points at cell centers. Iterate each point up to `maxiter=200` times and calculate the area. Measure runtime using a wall-clock timer (`omp_get_wtime()`).

Start with a serial version of the code, then parallelize it without automatic scheduling. You can do this by splitting the work for the outer loop equally among the different threads, while keeping the inner loop serial. You can use reduction for summing the number of points.

Next, create a second version in which you parallelize using automatic work sharing with static schedule.

In the final results, use `maxiter=2000`. Use an ipython notebook (or python script) to automatically evaluate the area for 1, 2, 4, 8, 16, 32 threads present a table with a) the area, b) the timing, c) the speedup, d) the parallel efficiency. Generate plots of timing, speedup and efficiency as function of the number of threads. Can you fit Amdahl's law and extrapolate to the maximum possible theoretical speedup? Can you repeat with smaller (or larger) N (and smaller/larger `maxiter` if required) and find the order of convergence of the method? Is it faster or slower than MCMC?

Present all your findings with proper explanations in a structured pdf report. Submit the report plus the source codes (c and python/ipython).